

Clean_Customer

Data Cleaning Steps & Observations

E-Commerce Sales Dataset

14-step cleaning documentation

Step 1 – Check Customer_ID values

Observation: Customer_ID values were reviewed and inconsistent formats such as C-1464, C-1486, C-1491 and C-1499 were identified.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

Paste

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

ChatGPT

J4

fx

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level					
		onilla	55	Male	Donaldton	North Carolina	USA						
		elson	26	Female	Phillipsmouth	North Dakota	USA	Platinum					
		n Price	29	Female	Kylebury	Alabama	USA	Platinum					
		arris	59	Female	Thomastown	Ohio	USA	Bronze					
		ollins	64	Female	Nicholasshire	California	USA	Silver					
		aughn	24	Male	East Amandamouth	Oregon	USA	Bronze					
		ackson	29	Female	Deborahborough	South Dakota	USA	Platinum					
		oyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze					
		oward	53	Female	Maynardton	New Hampshire	USA	Gold					
		Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum					
		Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze					
		pez	28	Female	East Krista	New York	USA	Gold					
		mbert	55	Female	North Kathryn	Kansas	USA	Platinum					
		andez	32	Female	New Becky	Montana	USA	Silver					
		ohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver					
		arez	49	Female	Lawsonhaven	Washington	USA	Platinum					
		ott	24	Female	South Christopher	Pennsylvania	USA	Bronze					
19	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum					
20	CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA						

Sort A to Z

Sort Z to A

Sort by Color

Sheet View

Clear Filter From "Customer_ID"

Filter by Color

Text Filters

Search

(Select All)

C-1464

C-1486

C-1491

C-1499

CUST1000

CUST1001

CUST1002

CUST1003

CUST1004

OK

Cancel

Customer_Dim

Product_Dim

Store_Dim

Sales_Fact

Clean_Customer

Clean_Product

Clean_Store

Clean_Sales

Ready

Accessibility: Investigate

30°C

Mostly cloudy

Search

ENG

IN

17:28

25-09-2026

Step 2 – Open Find & Replace for Customer_ID

Observation: Find & Replace was used to change the inconsistent ID format. Example: C-1464 → CUST1464.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

LB

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

Paste

Clipboard

Calibri11

Font

Wrap Text

Merge & Center

Alignment

General

Number

Conditional Formatting

Format as Table

Cell Styles

Styles

Insert

Delete

Format

Cells

AutoSum

Fill

Clear

Editing

Sort & Filter

Find & Select

Add-ins

ChatGPT

J4

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level					
466	C-1464	David Pierce	45	Female	Grayland	Washington	USA	Platinum					
488	C-1486	Eric Lee	22	Female	Lake April	Vermont	USA	Bronze					
493	C-1491	Kaitlin Brown	52	Female	Kellyberg	South Dakota	USA	Gold					
501	C-1499	Philip Pennington	27	Female	East Valeriewiew	Oregon	USA	Platinum					
502													
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Find and Replace

FindReplace

Find what:

C-1464

No Format Set

Format...

Replace with:

CUST1464

No Format Set

Format...

Within:

Sheet

Match case

Search:

By Rows

Match entire cell contents

Look in:

Formulas

Options <<

Replace All

Replace

Find All

Find Next

Close

Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_ProductClean_StoreClean_Sales

Ready4 of 500 records foundAccessibility: Investigate

30°C

Mostly cloudy

Search

ENG IN

17:29

25-09-2026

Step 3 – Complete Customer_ID replacement

Observation: Excel confirmed that the Customer_ID replacement was completed successfully.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

LB

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

Paste

Clipboard

Calibri11

B*I*U

Font

Align Center

Wrap Text

Merge & Center

General

Number

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Fill

Clear

Sort & Filter

Find & Select

Add-ins

ChatGPT

Find and Replace

Find what:

Replace with:

Within: Sheet

Search: By

Look in: Formulas

Replace All

Replace

Find All

Find Next

Close

Microsoft Excel

All done. We made 1 replacements.

OK

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level					
466	CUST1464	David Pierce	45	Female	Grayland	Washington	USA	Platinum					
488	C-1486	Eric Lee	22	Female	Lake April	Vermont	USA	Bronze					
493	C-1491	Kaitlin Brown	52	Female	Kellyberg	South Dakota	USA	Gold					
501	C-1499	Philip Pennington	27	Female	East Valeriew	Oregon	USA	Platinum					
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Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_ProductClean_StoreClean_Sales

Ready4 of 500 records foundAccessibility: Investigate

30°CMostly cloudy

Search

ENG IN

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Step 4 – Standardize Name capitalization

Observation: The Name column was cleaned using PROPER(). Example: =PROPER(B2).

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

Paste

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

ChatGPT

J2=PROPER(B2)

	A	B	C	D	E	F	G	H	I	J	K
	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level			
1	CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA			Sheena Bonilla	
2	CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum		Eugene Nelson	
3	CUST1002	dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum		Dr. Benjamin Price	
4	CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze		Jeremy Harris	
5	CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver		Caroline Collins	
6	CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze		Christine Vaughn	
7	CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum		Amber Jackson	
8	CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze		Daniel Boyd	
9	CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold		Melissa Howard	
10	CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum		Ms. Maria Leach	
11	CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze		Dominic Page	
12	CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold		Brian Lopez	
13	CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum		Kenneth Lambert	
14	CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver		Nathan Hernandez	
15	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver		Ashley Cohen	
16	CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum		Laura Alvarez	
17	CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze		John Scott	
18	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum		Adam Cruz	
19	CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA			Rhonda Moss	

<>

Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_ProductClean_StoreClean_Sales+

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Search

19:31 25-09-2026

Step 5 – Check Age values

Observation: The Age column was reviewed using Number Filters for blank, unusual or invalid values.

Ecommerce_Sales_Dataset_updated

File Home Insert Draw Page Layout Formulas Data Review View Help

- PivotTable Recommended PivotTables
- Shapes SmartArt Screenshot
- Pictures Icons 3D Models Illustrations
- Recommended Charts Charts Maps PivotChart Sparklines
- Line Column Win/Loss Filters Slicer Timeline Link Comment Text Equation Symbol

	A	B	C	D	E	F	G	H	I	J	K
	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level			
1	CUST1000			Male	Donaldton	North Carolina	USA				
2	CUST1001			Female	Phillipsmouth	North Dakota	USA	Platinum			
3	CUST1002			Female	Kylebury	Alabama	USA	Platinum			
4	CUST1003			Female	Thomastown	Ohio	USA	Bronze			
5	CUST1004			Female	Nicholasshire	California	USA	Silver			
6	CUST1005			Male	East Amandamouth	Oregon	USA	Bronze			
7	CUST1006			Female	Deborahborough	South Dakota	USA	Platinum			
8	CUST1007				Pamelafurt	West Virginia	USA	Bronze			
9	CUST1008				aynardton	New Hampshire	USA	Gold			
10	CUST1009				elissashire	Pennsylvania	USA	Platinum			
11	CUST1010				Michaelburgh	Pennsylvania	USA	Bronze			
12	CUST1011				ast Krista	New York	USA	Gold			
13	CUST1012				rth Kathryn	Kansas	USA	Platinum			
14	CUST1013				ew Becky	Montana	USA	Silver			
15	CUST1014				Kristamouth	Nebraska	USA	Silver			
16	CUST1015				wsonhaven	Washington	USA	Platinum			
17	CUST1016				n Christopher	Pennsylvania	USA	Bronze			
18	CUST1017				andonville	Georgia	USA	Platinum			
19	CUST1018	Rhonda Moss	43		uth William	Kansas	USA				

Sort & Filter - Age

- Sort Smallest to Largest
- Sort Largest to Smallest
- Sort by Color
- Sheet View
- Clear Filter From "Age"
- Filter by Color
- Number Filters

Search:

- (Select All)
- 18
- 19
- 20
- 21
- 22
- 23
- 24
- 25

Comparison Operators:

- Equals...
- Does Not Equal...
- Greater Than...
- Greater Than Or Equal To...
- Less Than...
- Less Than Or Equal To...
- Between...
- Top 10...
- Above Average
- Below Average
- Custom Filter...

Data Source: Customer Dim | Product Dim | Store Dim | Sales_Fact | Clean_Customer | Clean_Product | Clean_Store | Clean_Sales

Ready Accessibility: Investigate

29°C Mostly cloudy Search ENG IN 19:42 25-09-2026

Step 6 – Review Gender values

Observation: The Gender column was filtered to review Male and Female values.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

PivotTableRecommended PivotTables

Tables

PicturesShapesIcons3D Models

Illustrations

SmartArtScreenshot

Recommended Charts

Charts

MapsPivotChart

LineColumnWin/Loss

Sparklines

SlicerTimeline

Filters

Link

Links

Comment

Comments

Text

EquationSymbol

Symbols

K21

fx

	A	B	C	D	E	F	G	H	I	J	K
	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level			
1	CUST1000	Sheena Bonilla			Donaldton	North Carolina	USA				
2	CUST1001	Eugene Nelson			Phillipsmouth	North Dakota	USA	Platinum			
3	CUST1002	Dr. Benjamin Price			Kylebury	Alabama	USA	Platinum			
4	CUST1003	Jeremy Harris			Thomastown	Ohio	USA	Bronze			
5	CUST1004	Caroline Collins			Nicholasshire	California	USA	Silver			
6	CUST1005	Christine Vaughn			East Amandamouth	Oregon	USA	Bronze			
7	CUST1006	Amber Jackson			Deborahborough	South Dakota	USA	Platinum			
8	CUST1007	Daniel Boyd			West Pamelafurt	West Virginia	USA	Bronze			
9	CUST1008	Melissa Howard			Maynardton	New Hampshire	USA	Gold			
10	CUST1009	Ms. Maria Leach			Melissashire	Pennsylvania	USA	Platinum			
11	CUST1010	Dominic Page			Port Michaelburgh	Pennsylvania	USA	Bronze			
12	CUST1011	Brian Lopez			East Krista	New York	USA	Gold			
13	CUST1012	Kenneth Lambert			North Kathryn	Kansas	USA	Platinum			
14	CUST1013	Nathan Hernandez			New Becky	Montana	USA	Silver			
15	CUST1014	Ashley Cohen			Lake Kristamouth	Nebraska	USA	Silver			
16	CUST1015	Laura Alvarez			Lawsonhaven	Washington	USA	Platinum			
17	CUST1016	John Scott			South Christopher	Pennsylvania	USA	Bronze			
18	CUST1017	Adam Cruz			Brandonville	Georgia	USA	Platinum			
19	CUST1018	Rhonda Moss			South William	Kansas	USA				

Sort A to Z

Sort Z to A

Sort by Color

Sheet View

Clear Filter From "Gender"

Filter by Color

Text Filters

Search

☒ (Select All)

☒ Female

☒ Male

OK

Cancel

Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_ProductClean_StoreClean_Sales

ReadyAccessibility: Investigate

29°C Mostly cloudy

Search

ENG IN

19:46 25-09-2026

Step 7 – Validate Gender using ChatGPT for Excel

Observation: ChatGPT for Excel validated the Gender column. The result shown states that all 500 rows contain valid Male or Female values with no blanks or invalid entries.

Ecommerce_Sales_Dataset_updated • Saved to this PC

Search

Upgrade your plan

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpTable Design

Paste

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

ChatGPT

D2

Female

	A	B	C	D	E	F	G	H
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty
2	CUST1000	Sheena Bonilla	55	Female	Donaldton	North Carolina	USA	Unkr
3	CUST1001	Eugene Nelson	26	Male	Phillipsmouth	North Dakota	USA	Plati
4	CUST1002	Dr. Benjamin Price	29	Male	Kylebury	Alabama	USA	Plati
5	CUST1003	Jeremy Harris	59	Male	Thomastown	Ohio	USA	Bro
6	CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Sil
7	CUST1005	Christine Vaughn	24	Female	East Amandamouth	Oregon	USA	Bro
8	CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Plati
9	CUST1007	Daniel Boyd	29	Male	West Pamelafurt	West Virginia	USA	Bro
10	CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gr
11	CUST1009	Ms. Maria Leach	64	Female	Melissashire	Pennsylvania	USA	Plati
12	CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bro
13	CUST1011	Brian Lopez	28	Male	East Krista	New York	USA	Gr
14	CUST1012	Kenneth Lambert	55	Male	North Kathryn	Kansas	USA	Plati
15	CUST1013	Nathan Hernandez	32	Male	New Becky	Montana	USA	Sil
16	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Sil
17	CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Plati
18	CUST1016	John Scott	24	Male	South Christopher	Pennsylvania	USA	Bro
19	CUST1017	Adam Cruz	60	Male	Brandonville	Georgia	USA	Plati
20	CUST1018	Rhonda Moss	43	Female	South William	Kansas	USA	Unkr

Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_Prc

Count: 500

ChatGPT

Validate Customer Name an...

- Verified all 500 rows contain only valid Male or Female values, with no blanks or invalid entries. [Clean_Customer D2:D501](#)
- Spot-checked the beginning and end of the dataset after the update. [Clean_Customer B2:D25](#)
[Clean_Customer B494:D501](#)

Clean_Customer D2:D501 selected

Ask anything, @ for context

GPT-5.6 Luna

28°C Mostly cloudy

Search

ENG IN

20:23 25-09-2026

Step 8 – Check City values

Observation: The City column was reviewed using the filter to check for blank or inconsistent entries.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

LB

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

PivotTableRecommended PivotTablesTable

PicturesShapesIcons3D ModelsSmartArtScreenshot

Recommended Charts

Charts

MapsPivotChart

Sparklines

Filters

Links

Comments

Text

Symbols

J2

Customer_ID

Name

Age

Gender

City

State

Country

Loyalty_Level

1	CUST1000	Sheena Bonilla	55		North Carolina	USA		
2	CUST1005	Christine Vaughn	24		Oregon	USA	Bronze	
7	CUST1009	Ms. Maria Leach	64		Pennsylvania	USA	Platinum	
11	CUST1010	Dominic Page	53		Pennsylvania	USA	Bronze	
12	CUST1018	Rhonda Moss	43		Kansas	USA		
20	CUST1020	Bridget Allen	59		Delaware	USA	Silver	
22	CUST1024	Heather Collins	40		Idaho	USA	Platinum	
26	CUST1027	Leslie Johnson	31		Utah	USA		
29	CUST1031	Sarah Vance	41		Oregon	USA	Silver	
33	CUST1033	Robert Jones Dds	25		Oklahoma	USA	Platinum	
35	CUST1035	Rachel Myers	52		North Dakota	USA	Silver	
37	CUST1040	Eric Vega	50		Kansas	USA	Gold	
42	CUST1041	James Gibbs	21		Alaska	USA	Platinum	
43	CUST1042	William Martin	19		Connecticut	USA	Platinum	
44	CUST1043	Joseph Snow	45		Vermont	USA	Gold	
45	CUST1044	Denise Williams	34		Hawaii	USA	Bronze	
46	CUST1045	Jordan Jackson Dds	54		Arizona	USA	Gold	
47	CUST1048	April Mullins	54		North Carolina	USA	Platinum	
50	CUST1052	Debra Liu	55	Male	South Timtown	Minnesota	USA	Bronze

Sort A to Z

Sort Z to A

Sort by Color

Sheet View

Clear Filter From "City"

Filter by Color

Text Filters

Search

West Markton

West Mathewside

West Rebecca

West Sarah

West Theodorechester

West Tylerfort

West Victoriaborough

Williamsonland

Wongberg

OK

Cancel

Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_ProductClean_StoreClean_Sales

Ready

Accessibility: Investigate

29°C Mostly cloudy

Search

ENG IN

19:52 25-09-2026

Observation: The State column was reviewed using the filter to check for missing or inconsistent entries.

Observation: The State column was reviewed using the filter to check for missing or inconsistent entries.

[illegible]

Step 10 – Check Country values

Observation: The Country column showed United States of America and USA, so the values were identified for standardization.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpTable Design

Paste

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Align Center

Wrap Text

General

Number

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

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AutoSum

Fill

Clear

Sort & Filter

Find & Select

Add-ins

ChatGPT

H2

Customer_ID

Name

Age

Gender

City

State

Country

Loyalty_Level

1	CUST1000	sheena bonilla	55	Male	Donaldton									
2	CUST1001	Eugene Nelson	26	Female	Phillipsmouth		Platinum							
3	CUST1002	dr. Benjamin Price	29	Female	Kylebury		Platinum							
4	CUST1003	jeremy harris	59	Female	Thomastown		Bronze							
5	CUST1004	Caroline Collins	64	Female	Nicholasshire		Silver							
6	CUST1005	Christine Vaughn	24	Male	East Amandamouth		Bronze							
7	CUST1006	Amber Jackson	29	Female	Deborahborough		Platinum							
8	CUST1007	Daniel Boyd	29	Female	West Pamelafurt		Bronze							
9	CUST1008	Melissa Howard	53	Female	Maynardton		Gold							
10	CUST1009	Ms. Maria Leach	64	Male	Melissashire		Platinum							
11	CUST1010	Dominic Page	53	Male	Port Michaelburgh		Bronze							
12	CUST1011	Brian Lopez	28	Female	East Krista		Gold							
13	CUST1012	Kenneth Lambert	55	Female	North Kathryn		Platinum							
14	CUST1013	Nathan Hernandez	32	Female	New Becky		Silver							
15	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth		Silver							
16	CUST1015	Laura Alvarez	49	Female	Lawsonhaven		Platinum							
17	CUST1016	John Scott	24	Female	South Christopher		Bronze							
18	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA							
19	CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA							

Sort A to Z

Sort Z to A

Sort by Color

Sheet View

Clear Filter From "Country"

Filter by Color

Text Filters

Search

(Select All)

United States of America

USA

OK

Cancel

Customer_Dim

Product_Dim

Store_Dim

Sales_Fact

Clean_Customer

Clean_Product

Clean_Store

Clean_Sales

Ready

Accessibility: Investigate

30°C

Mostly cloudy

Search

ENG IN

17:18

25-09-2026

Step 11 – Standardize Country values

Observation: Find & Replace was used to change United States of America to USA.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

LB

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpTable Design

Paste

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Calibri11A⁺A⁻

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Conditional Formatting

Format as Table

Cell Styles

Styles

Insert

Delete

Format

Cells

Editing

AutoSum

Fill

Clear

Sort & Filter

Find & Select

Add-ins

ChatGPT

Comments

Share

G132

United States of America

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level					
132	CUST1130	Jamie Reyes	29	Male	jonesbury	Connecticut	United States of America	Platinum					
143	CUST1141	Cristian Wright	30	Male	Port Denise	California	United States of America	Bronze					
164	CUST1162	Mariah Brown	37	Male	Palmerbury	Tennessee	United States of America	Silver					
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Find and Replace

Find

Replace

Find what:United States of America

No Format Set

Format...

Replace with:USA

No Format Set

Format...

Within:Sheet

Match case

Search:By Rows

Match entire cell contents

Look in:Formulas

Options <<

Replace All

Replace

Find All

Find Next

Close

Customer_Dim

Product_Dim

Store_Dim

Sales_Fact

Clean_Customer

Clean_Product

Clean_Store

Clean_Sales

Edit

Filter Mode

Accessibility: Investigate

100%

30°C

Mostly cloudy

Search

ENG IN

17:22

25-09-2026

Step 12 – Confirm Country replacement

Observation: Excel confirmed that 3 Country replacements were made.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpTable Design

Paste

Clipboard

Calibri11

B*I*U

Font

Align Center

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Fill

Clear

Sort & Filter

Find & Select

Add-ins

ChatGPT

Editing

Add-ins

ChatGPT

USA

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level					
132	CUST1130	Jamie Reyes	29	Male	jonesbury	Connecticut	USA	Platinum					
143	CUST1141	Cristian Wright	30	Male	Port Denise	California	USA	Bronze					
164	CUST1162	Mariah Brown	37	Male	Palmerbury	Tennessee	USA	Silver					
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Find and Replace

Find

Find what:

Replace with:

Within:

Search:

Look in:

Replace All

Replace

Find All

Find Next

Close

Microsoft Excel

All done. We made 3 replacements.

OK

Customer_Dim

Product_Dim

Store_Dim

Sales_Fact

Clean_Customer

Clean_Product

Clean_Store

Clean_Sales

ReadyFilter ModeAccessibility: Investigate

30°C

Mostly cloudy

Search

ENG IN

17:22

25-09-2026

Step 13 – Find missing Loyalty_Level values

Observation: The Loyalty_Level filter was used to display blank values. Four missing records were identified.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

PivotTableRecommended PivotTablesTablePicturesShapesIconsSmartArtScreenshot3D ModelsIllustrations

Recommended ChartsChartsMapsPivotChartSparklinesLineColumnWin/LossFiltersTimelineLinkCommentTextEquationSymbol

J2fx

	A	B	C	D	E	F	G	H	I	J	K
	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level			
1	CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota					
3	CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama					
4	CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio					
5	CUST1004	Caroline Collins	64	Female	Nicholasshire	California					
6	CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota					
8	CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia					
9	CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire					
10	CUST1011	Brian Lopez	28	Female	East Krista	New York					
13	CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas					
14	CUST1013	Nathan Hernandez	32	Female	New Becky	Montana					
15	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska					
16	CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington					
17	CUST1016	John Scott	24	Female	South Christopher	Pennsylvania					
18	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia					
19	CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey					
21	CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma					
23	CUST1022	Paul Fox	33	Female	Michaelton	Hawaii					
24	CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon					
25	CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Silver			
27								Bronze			

Sort A to Z

Sort Z to A

Sort by Color

Sheet View

Clear Filter From "Loyalty_Level"

Filter by Color

Text Filters

Search

(Select All)

Bronze

Gold

Platinum

Silver

(Blanks)

OKCancel

<>

Customer_DimProduct_DimStore_DimSales_FactClean_CustomerClean_ProductClean_StoreClean_Sales

+<=>100%

Ready 262 of 500 records foundAccessibility: Investigate

29°C Mostly cloudy

Search

ENG IN

19:57 25-09-2026

Step 14 – Fill missing Loyalty_Level values

Observation: The four missing Loyalty_Level values were filled with Unknown.

Ecommerce_Sales_Dataset_updated

Search

Upgrade your plan

Comments

Share

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpTable Design

Paste

Clipboard

Calibri11

B*I*U

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

ChatGPT

B5

Jeremy Harris

	A	B	C	D	E	F	G	H	I	J	K
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level			
2	CUST1000	Sheena Bonilla	55	Female	Donaldton	North Carolina	USA	Unknown			
20	CUST1018	Rhonda Moss	43	Female	South William	Kansas	USA	Unknown			
29	CUST1027	Leslie Johnson	31	Female	Mollyshire	Utah	USA	Unknown			
39	CUST1037	Christopher Carter	27	Male	Brendastad	South Dakota	USA	Unknown			
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Product_DimStore_DimSales_FactClean_CustomerClean_Customer – ObservationClean_ProductClean_Store

Ready4 of 500 records foundAccessibility: Investigate

27°C Mostly cloudy

Search

ENG IN

22:02 25-09-2026

Final Clean_Customer Observation

- Customer_ID values were standardized where inconsistent formats were identified.
- Names were standardized using PROPER().
- Age values were reviewed using Number Filters.
- Gender values were validated as Male/Female using ChatGPT for Excel.
- City and State columns were checked using filters.
- Country values were standardized from United States of America to USA.
- Four missing Loyalty_Level values were filled with Unknown.
- VLOOKUP, PivotTable and charts were not required for this cleaning section.